Microsoft - InnovationChallengeMarch 2026

AccessMesh-AI

REDEFINING UNIVERSAL, MULTIMODAL COMMUNICATION FOR AN INCLUSIVE FUTURE.

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BREAKING BARRIERS.
BUILDING BRIDGES.

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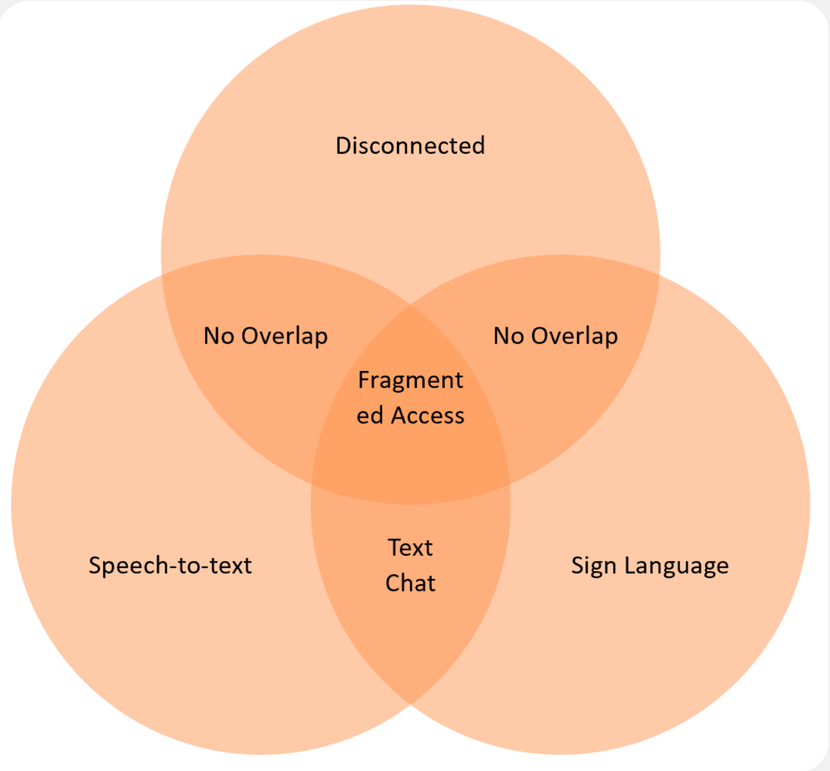
The Problem

Fragmented Tools. Excluded Users.

430M+ PEOPLE WORLDWIDE ARE LEFT BEHIND.



Why Current Solutions Fail

Captioning	Not collaborative
Translation	Not real-time
Sign Language	Not integrated
Meeting Tools	Not inclusive

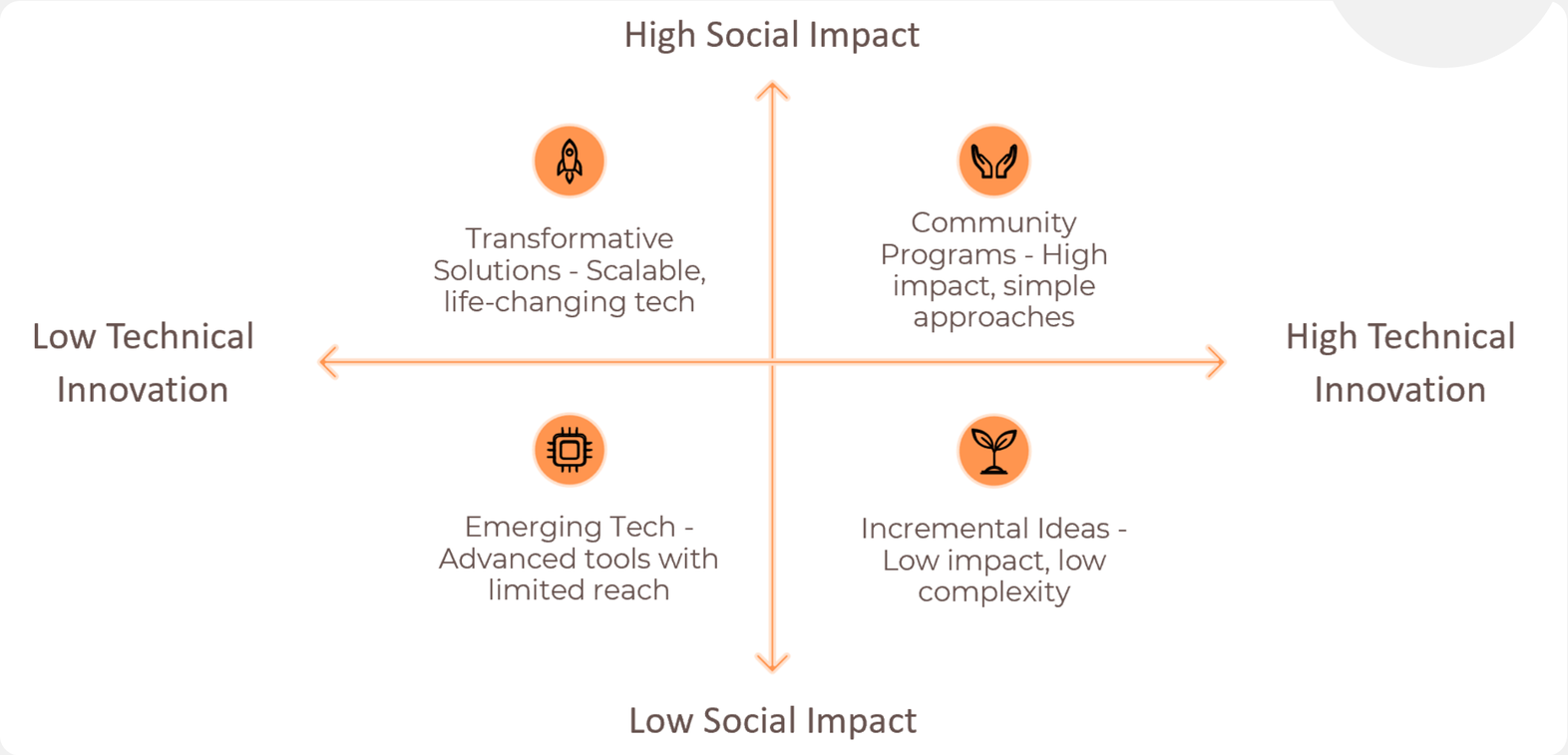
Innovation With Purpose

One conversation, multiple modalities, zero barriers.

VOICE. TEXT. GESTURE.
ALL CONNECTED.

Intelligent Processing - AI agents adapt communication across formats in real-time.

Seamless Multimodal Delivery - Instant translation, subtitles, and sign language for everyone.



INSTANT CRYSTAL-CLEAR TRANSCRIPTION

VOICE INPUT

SEAMLESS MULTILINGUAL TRANSLATION

TEXT INPUT

REAL-TIME SIGN LANGUAGE INTERPRETATION

GESTURE INPUT

AccessMesh-AI

Real-Time Accessibility in Action

Video Slide

Intelligent Agents. Real-Time Collaboration.

RouterAgent
Directs downstream flow

AccessibilityAgent
Generates metadata

TranslationAgent
Multilingual processing

SummaryAgent
Meeting documentation

AvatarAgent
Gloss rendering

HOW IT WORKS

USER INPUT - SPEECH, TEXT, OR GESTURES CAPTURED IN REAL TIME

↓

AI PIPELINE - PROCESSING THROUGH SPECIALIZED AGENTS

↓

ENRICHED OUTPUT - TRANSCRIPTION, TRANSLATION, ACCESSIBILITY FEATURES

↓

BROADCAST - REAL-TIME DELIVERY TO ALL PARTICIPANTS

Event-Driven Agent Mesh

Architeture

- Frontend
- BackendAPI
- MCP Layer
- Agent Mesh
- Azure Cloud

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BUILT ON AZURE SERVICES

Built on Azure Services

🟦 **1. INTELLIGENCE LAYER (IA COGNITIVA)**

- AZURE OPENAI → REASONING & ROUTING
- AZURE AI SPEECH → PERCEPTION
- TRANSLATOR → LINGUISTIC BRIDGE

🟪 **2. REAL-TIME LAYER**

- AZURE WEB PUBSUB
- AZURE SERVICE BUS

🟧 **3. STATE & MEMORY**

- AZURE COSMOS DB

🟩 **4. TRUST & SAFETY LAYER**

- AZURE CONTENT SAFETY
- AZURE ENTRA ID
- AZURE KEY VAULT

🟥 **5. OBSERVABILITY LAYER**

- AZURE APPLICATION INSIGHTS

(INSTRUMENTED VIA OPENTELEMETRY)

Enterprise-Ready. Deployed. Scalable.

Five Pillars of Production Maturity

Full-Stack Implementation
React and FastAPI architecture

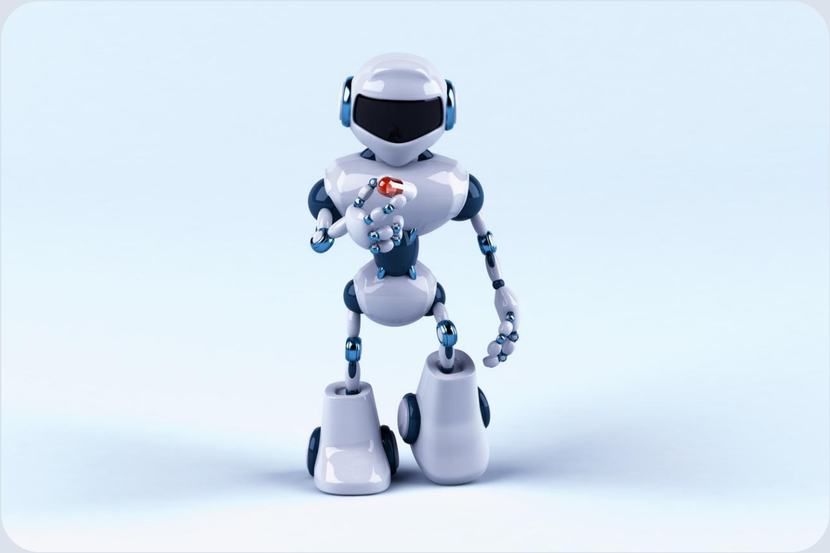
Real Database
Cosmos DB for persistence

Enterprise Security
Key Vault and JWT

Graceful Fallbacks
Resilient offline operation

Scalable Infrastructure
Real-time Web PubSub

Responsible AI Design



🛡️ **Safety**
Azure Content Safety moderate messages before broadcast.

👁️ **Transparency**
Users see:
- transcription
- translation
- gloss adaptation

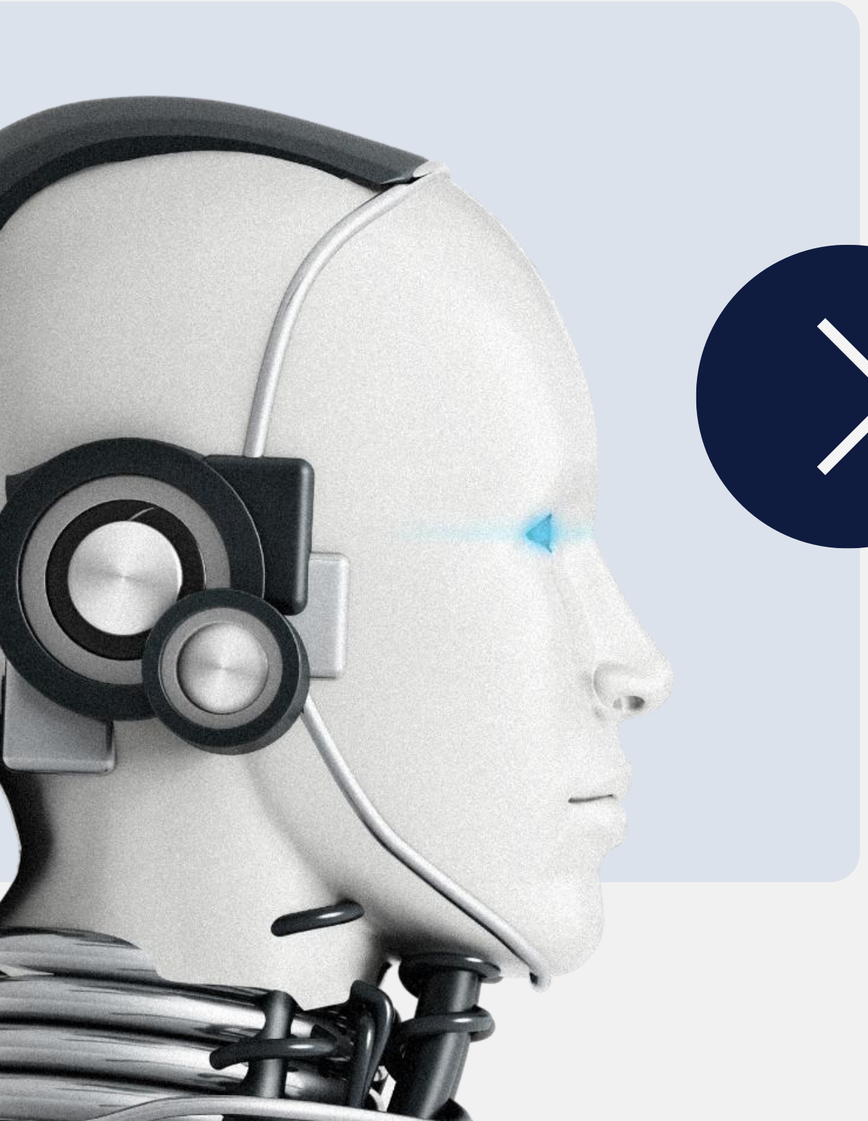
👷 **Human Control**
User chooses:
- modality preference
- communication mode

🔒 **Privacy by Architecture**
Most process = streaming transient.

Input
↓
Content Safety (pre-processing)
↓
Agent Processing
↓
Transparency Layer
↓
User Control
↓
Broadcast

Data privacy risks

WHEN HANDLING SENSITIVE INFORMATION



Impact

- **430M+ People with disabling hearing loss:** Gain real-time access to spoken conversations through live transcription, sign adaptation, and AI-powered speech synthesis.
- **1.3B+ People living with visual impairment:** Can participate through multimodal communication — voice, audio feedback, and accessible interaction channels.
- **16% of the global population lives with a disability**: AccessMesh-AI transforms communication into an inclusive, real-time experience for everyone.

From MVP to Market

1

Phase 1 (Now)

Executing MVP and validating core concepts with real users.

2

Phase 2 (Q2 2026)

Beta launch with key enterprise partners for rigorous feedback.

3

Phase 3 (Q4 2026)

Public launch and deployment of developer API integrations.

4

Phase 4 (2027)

Global expansion across 50+ languages with full mobile accessibility.

Thank you